

PURE Power Home Systems - Complete Comparison Guide

Introduction

PURE is a company incubated out of I-TIC, IIT Hyderabad, committed to driving the future of energy storage and electric mobility. Their PuREPower Home series offers advanced power backup systems designed to run high-power appliances such as air conditioners, geysers, mixers, and coolers.

This guide provides a detailed comparison of the four models in the PuREPower Home series and recommendations for which model best suits different use cases.

Comparison Table

Specification	PuRE Power Home 3.0	PuREPower Home 5.0	PuREPower Home 12.0	PuREPower Home 20.0
Basic Specifications				
Capacity	3 KVA/kWh	5 KVA/kWh	12 KVA/kWh	20 KVA
Backup Time	Up to 6 hours	Up to 8 hours	Up to 12 hours	Up to 18 hours
Price	\$1,191 USD	\$1,589 USD	\$2,649.56 USD	\$4,967.95 USD
Physical Characteristics				
Dimensions (W × D × H)	407 × 621 × 128 mm	485 × 655 × 152 mm	485 × 1110 × 152 mm	970 × 2220 × 152 mm
Installation Style	Wall-mounted	Wall-mounted	Wall-mounted	Floor standing
Weight	39 kg	40 kg	82 kg	115 kg
Power Specifications				
Rated Output Power	3 KVA	5 KVA	12000W	20000W
Surge Power	6000 VA	11000 VA	22000 VA	36000 VA
Input Voltage	230 VAC	230 VAC	230 VAC	3 × 415 VAC
Efficiency (Peak)	97%	>94%	>94%	>97%
Solar & Charging				
Solar Charger Type	MPPT	MPPT	MPPT	MPPT
Number of MPPTs	1	1	2	3
Maximum PV Array Power	3000W	7500W	5500W × 2	7500W × 3
Maximum PV Open Circuit Voltage	450 VDC	500 VDC	500 VDC	500 VDC
Battery Specifications				
Battery Technology	Lithium Ion	Lithium Ion	Lithium Ion	Lithium Ion
Number of Batteries	1	1	2	4
Nominal Voltage	48V	48V	48V	48V
Battery Capacity	2.5 kW	5 kW	12 kW	20 kW
Features				
Smart AI	Yes	Yes	Yes	Yes
Solar Support	Yes	Yes	Yes	Yes

Specification	PuRE Power Home 3.0	PuREPower Home 5.0	PuREPower Home 12.0	PuREPower Home 20.0
Advanced Thermal Management	Yes	Yes	Yes	Yes
5th Gen Power Electronics	Yes	Yes	Yes	Yes
Operating Temperature Range	-10°C ~ 60°C	-10°C ~ 60°C	-10°C ~ 60°C	-10°C ~ 60°C
Warranty				
Standard Warranty	5 years	5 years	5 years	5 years
Extended Warranty	12 years	12 years	12 years	12 years
Communication & Interface				
Communication Interface	BMS, Dry-contact, CAN, Bluetooth, WiFi, AI Cloud	BMS, Dry-contact, CAN, Bluetooth, WiFi, AI Cloud	BMS, Dry-contact, CAN, Bluetooth, WiFi, AI Cloud	BMS, Dry-contact, CAN, Bluetooth, WiFi, AI Cloud
IP Protection	IP65	IP65	IP65	IP65

Note: All systems include features like advanced thermal management systems with Nano PCM technology, pure sine wave output, and multiple protection mechanisms including 5th Gen BMS.

Best Use Cases by Model

PuRE Power Home 3.0 (\$1,191 USD)

Best for:

- Small apartments or homes (1-2 bedrooms)
- Basic power backup needs
- Running essential appliances during short outages
- Budget-conscious consumers
- Homes with minimal power requirements (around 2-3 kW peak load)
- Backup for a few critical devices like lights, fans, refrigerator, and perhaps one small AC

PuREPower Home 5.0 (\$1,589 USD)

Best for:

- Medium-sized homes (2-3 bedrooms)

- Running multiple appliances simultaneously
- Longer outage coverage for essential items
- Small home offices that need consistent power
- Homes with moderate power requirements (around 3-5 kW peak load)
- Good balance of price and performance for most residential needs

PuREPower Home 12.0 (\$2,649.56 USD)

Best for:

- Larger homes (3-4+ bedrooms)
- Running multiple high-power appliances simultaneously
- Home offices with servers or specialized equipment
- Homes with substantial power requirements (8-10 kW peak load)
- Families that need to maintain normal lifestyle during extended outages
- Small businesses or shops requiring reliable backup power

PuREPower Home 20.0 (\$4,967.95 USD)

Best for:

- Large homes or small commercial spaces
- Homes with 3-phase equipment like residential elevators
- Full home backup with no compromises during extended outages
- Properties with very high power requirements (15+ kW peak load)
- Multiple AC units running simultaneously
- Small businesses that cannot afford power interruptions
- Remote locations where outages may be frequent and prolonged

Recommendations for Selecting the Right System

1. **Assess your power needs:** Before choosing a model, calculate your total power requirements. Make a list of appliances you need to run during outages (especially high-draw items like ACs, geysers, and refrigerators) and their power consumption.
2. **Consider future expansion:** The systems allow for parallel connection (up to 15 units), so you could start with a smaller unit and expand later if needed.

3. **Solar integration potential:** All models support solar input, but with different capacities. If you plan to add solar panels, ensure the model you choose has adequate MPPT capacity for your planned solar array size.
4. **Installation requirements:** Note that the 3.0, 5.0, and 12.0 models are wall-mounted, while the 20.0 is floor-standing. Ensure you have appropriate space and structural support.
5. **Cost-benefit analysis:** The price jump from the 5.0 (\$1,589) to the 12.0 (\$2,649.56) model is significant. For residential use, the 5.0 model might offer the best balance of capacity and cost unless you have specific high power needs.
6. **Three-phase consideration:** Only the 20.0 model supports three-phase power (415 VAC). If you need to run a residential elevator or three-phase equipment, this is your only option.
7. **Extended warranty:** Consider the value of the extended warranty (12 years vs. standard 5 years).
8. **Monitoring needs:** All systems offer smart features with Bluetooth, WiFi, and AI Cloud connectivity. If remote monitoring is important to you, ensure your internet connection is reliable.

Key Features Across All Models

Advanced Technology

- **Smart AI Integration:** Intelligent monitoring and management systems
- **5th Generation Power Electronics:** Maximum efficiency and reliability
- **Pure Sine Wave Output:** Clean power suitable for sensitive electronics

Thermal Management & Protection

- **Advanced Nano PCM Technology:** Superior heat dissipation for longer battery life
- **Wide Operating Temperature Range:** Functioning from -10°C to 60°C
- **Comprehensive Safety Features:** Overload, short circuit, and temperature protection
- **IP65 Rating:** Dust-tight and protected against water jets

Connectivity

- **Multiple Communication Options:** BMS, CAN, Bluetooth, WiFi and AI Cloud
- **Remote Monitoring Capabilities:** Track system performance and status remotely

Conclusion

The PURE Power Home series offers solutions to meet various power backup needs, from basic home backup to comprehensive power management for larger properties and businesses. When selecting the

right model, consider your current and future power requirements, budget constraints, and installation capabilities.

The sweet spot for most residential users will likely be either the 5.0 or 12.0 model, depending on home size and power requirements. The 3.0 is most economical but somewhat limited in capacity, while the 20.0 represents a significant investment that's primarily justified if you have three-phase equipment or exceptionally high power needs.

For optimal performance and longevity, proper installation by qualified technicians is recommended, along with regular maintenance according to the manufacturer's guidelines.

Document prepared based on PURE Power Home product specifications. All specifications and prices are current as of May 2025.